Build a Task Tracker

Mini Project | Week 1, Day 2 | Student brief and submission instructions

Goal: Build a menu-based Python Task Tracker that stores tasks in a JSON file and can reload the saved tasks when run again. Work in Google Colab, then submit your code through GitHub.

Required functions and program behaviour

Feature	Requirement
Add	Ask for a task title and store it as Pending.
View	Show a numbered task list with Pending or Done.
Complete	Select a task number and mark its done field True.
Delete	Select a task number and remove that task.
Save and Load	Save list to tasks.json with json.dump(); load with json.load() at startup.
Menu	Keep showing choices until the user selects Exit.
Error handling	Handle empty titles, invalid menu choices, invalid task numbers, and a missing JSON file.

Step-by-step build plan

- Create tasks = [] and store each task as {"title": "...", "done": False}.
- Write small functions: load_tasks(), save_tasks(), add_task(), view_tasks(), complete_task(), delete_task().
- Add a menu using while True and if/elif/else.
- After Add/Complete/Delete, save the updated list in tasks.json.
- At program start, try loading tasks.json; if missing, begin with an empty list.
- Test a blank title, invalid number and relaunch in the same Colab runtime.
- Export a Python script as mini_task_tracker.py and create README.md.
- Create a GitHub repository, upload project files, commit changes twice and submit the repository link.

Sample user interaction

```
--- TASK TRACKER ---
1. Add  2. View  3. Complete  4. Delete  5. Exit
Choice: 1
Task title: Read notes
Task added.
Choice: 2
1. Read notes [Pending]
Choice: 3
Task number: 1
Task completed.
Choice: 2
1. Read notes [Done]
```

Files to submit

- mini_task_tracker.py - complete Python program.
- README.md - project goal, features, and how to run the program.
- GitHub repository link, showing at least two meaningful commits.
- Be prepared for an individual working demonstration.

Google Colab reminder: files in a hosted runtime are temporary. For permanent storage, download tasks.json or save it to Google Drive; the classroom persistence test should reuse the same active runtime.

References for the workflow

GitHub file uploads:
https://docs.github.com/en/repositories/working-with-files/managing-files/adding-a-file-to-a-repository

Google Colab FAQ: https://research.google.com/colaboratory/faq.html